# JUNQI LU

GitHub | Google Scholar | Personal Website | Linkedin
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Luxembourg

## EDUCATION

**University of Luxembourg** Belval, Luxembourg

- **M.Sc. in Mathematics** Sep. 2026 – Jun. 2028 (Expected)
  - Mathematical Modelling and Computational Sciences track

**Beijing Institute of Technology** Beijing, China

- **B.S. in Computer Science** (Second Bachelor's Degree) Sep. 2024 – Jun. 2026
- **B.S. in Mathematics and Applied Mathematics** Oct. 2020 – Jun. 2024

## PUBLICATIONS

*(*: Equal contribution; #: Corresponding author)*

[1] **Junqi Lu***, Ruixiang Sun*, Xin Li#, Gaopeng Peng, Ao Zhang, and Mingzhong Wang. Prevailing Bisimulation Metric Learning Is Biased: Implicit Regularization and Its Remedy. In *Advances in Neural Information Processing Systems* (**NeurIPS**), 2026.

[2] **Junqi Lu**, Bosen Liu, Cuicui Pei, Qingan Qiu#, and Li Yang. Learning to optimize termination decisions under hybrid uncertainty of system lifetime and task duration. *Computers & Industrial Engineering*, 206, 111208, 2025.

[3] **Junqi Lu**, Qingan Qiu#, Rongchi Sun, and Yike Zhang. Simplicity Wins: How Myopic Decisions Enable Safe Learning in Mission-Critical Systems. (**Under Review**)

## RESEARCH EXPERIENCE

**Deep Reinforcement Learning & State Representation** Sep. 2025 – Jun. 2026
*Research Assistant* | Advisor: Prof. Xin Li | 1029 DRL Lab | Beijing Institute of Technology

- **Theory:** Identified double-sampling bias in deep bisimulation metric learning; showed via Itô SDE analysis how transition noise induces implicit Jacobian regularization and representation collapse.
- **Method:** Developed BSPO, a Fenchel-dual saddle-point formulation with an auxiliary network that estimates expected Bellman residuals using single-transition samples.
- **Experiments:** Evaluated BSPO on stochastic/visually distracting DeepMind Control tasks and six Atari games; the work led to a **NeurIPS 2026 co-first-author paper**.

**Reliability Engineering & Stochastic Control** Jun. 2023 – Jun. 2025
*Research Assistant* | Advisor: Prof. Qingan Qiu | Beijing Institute of Technology

- **Model:** Formulated Bayesian MDPs for sequential task termination under uncertain system lifetimes and task durations, with Gamma–exponential updates from censored observations.
- **Theory:** Proved existence and monotonicity of optimal termination thresholds and structural monotonicity of value functions.
- **Algorithm:** Designed an $O(N)$ monotone threshold-update algorithm; numerical experiments showed up to 45.11% lower cost than a non-termination policy. First-author paper in *Computers & Industrial Engineering* (2025).

## SELECTED PROJECTS

**Nitrogen Cycling Dynamics Model (MCM-ICM 2025 Finalist)** [GitHub: 100+ Stars]

- **Modeling:** Built a dynamic ecosystem model using modified Lotka–Volterra equations to study nitrogen flows across forest and agricultural food webs.
- **Reproducibility:** Released an open-source implementation with documentation and tutorials; the repository received **100+ GitHub stars**.
- **Recognition:** Earned **ICM Finalist (Top 2%)** among 27,456 participating teams.

**Formal Verification of a Mahjong Winning-Hand Checker** [Demo] [Code]
*AI4Math Summer School, Zhejiang University* Summer 2026

- **Implementation:** Independently specified and implemented a three-suit Mahjong winning-hand checker in **Lean 4**, covering standard hands, seven pairs, declared melds and kongs.
- **Formal verification:** Proved soundness and completeness of the executable checker against its declarative specification for every valid hand state.

## HONORS & AWARDS

- **Meritorious Winner**, Mathematical Contest in Modeling (MCM) 2023, 2026
- **Outstanding Student Award** (Top 10% university-wide), BIT Nov. 2025
- **Academic Excellence Scholarship** (Rank 1st in CS 2nd Bachelor Program), BIT Oct. 2025
- **Finalist (Top 2%)**, Interdisciplinary Contest in Modeling (ICM) Feb. 2025
- **University "Excellent Thesis" Prize**, BIT Jun. 2024
- **Second Prize**, China Undergraduate Mathematical Contest in Modeling (CUMCM) Nov. 2022

## SKILLS

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|---|---|
| **Research** | Reinforcement Learning, Representation Learning, Bayesian MDPs, Stochastic Control, Mathematical Optimization, SDE Analysis |
| **Programming** | Python (PyTorch, NumPy, Pandas), C++, SQL, MATLAB, Bash |
| **Tools** | Linux, Git, Docker, LaTeX, Make, Lean 4 (Formal Verification) |
| **Languages** | Chinese (Native), English (B2), German (A1) |

## OUTREACH & INTERESTS

**Technical Writing:** CNBlogs — 20+ posts on algorithms and programming, 22K+ views.
**Outside Research:** Distance running (half-marathon PB: 1:41:34); guitarist and bassist in university rock bands.